SCATS · Level 3 · Mastery

Five new ideas for SCATS, tested honestly

Connected cars, a learning agent, max-pressure, broken detectors and signal prediction: what each one does, how to test it fairly, and what really happened in SUMO.

smrahmanbd.github.io/scats-learning-lab

6 ideas, each building on the last

1 How to evaluate a new SCATS idea

3 seeds minimum to see whether a difference is real

2 Degree of saturation from connected vehicles

0.65 mean DS error with 5% connected vehicles

3 Learning split selection inside the SCATS envelope

+4.2 s RL delay penalty against SCATS, 3 seeds

4 SCATS and max-pressure

57.4 s bus delay under pure max-pressure (SCATS 29.7 s)

5 Resilience to detector faults

+60% delay increase with 30% of loops faulty and no alarms

6 Predicting signal timing for connected vehicles

2.63 s mean error of the planned-green predictor

Three surprises to start

Two of the five new ideas lost.

A reinforcement-learning agent and a max-pressure hybrid both gave more delay than standard SCATS on this lab's corridor. Reporting that honestly is what makes the other results believable.

Three broken loops in ten raised delay by 60%.

When 30% of the detectors failed, mean delay jumped from 36.9 to 59.2 seconds per vehicle. Two simple alarm rules brought it back to 40.7 s.

A connected car can mislead you by being lucky.

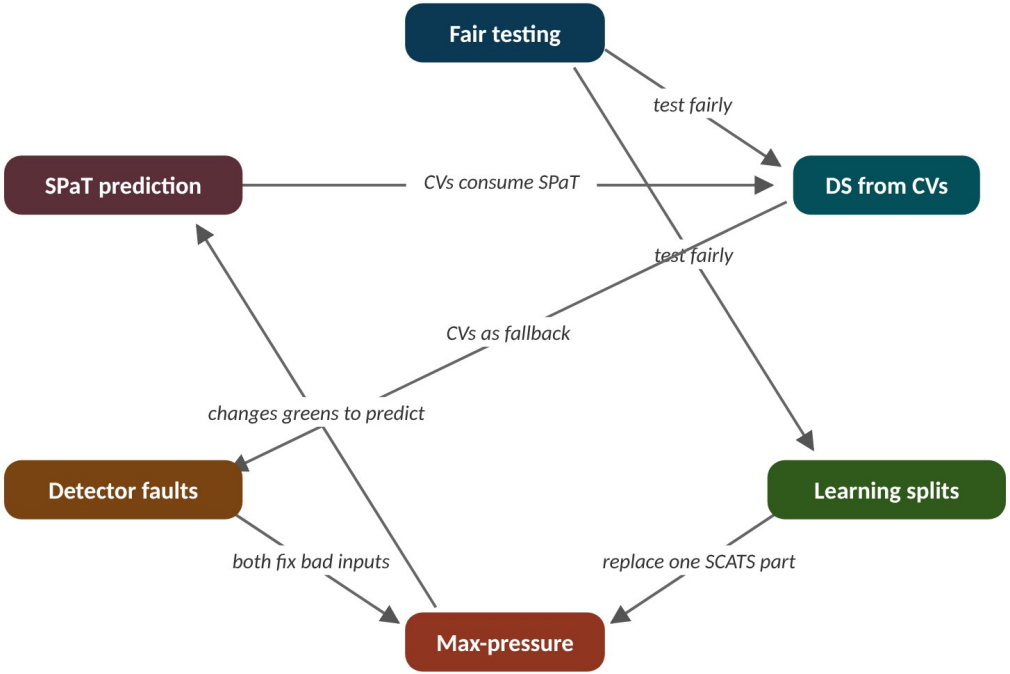
If only 1 car in 20 reports its position, one green may contain zero reporting cars and the next may contain three. With 5% penetration the DS estimate was off by 0.65 on average: bigger than many real DS values.

Words you will meet

Degree of saturation (DS)	How fully a green was used: 1.0 means the green was busy from start to end. SCATS measures it at stop-line loops.
Split plan	A stored way of sharing the cycle between phases, for example 'main road 55%, side road 45%'.
Baseline	The method you compare against, for example fixed time, actuated, or standard SCATS.
Seed	A number that fixes the random traffic in a simulation. Different seeds give different, equally likely days.
Connected vehicle (CV)	A car that reports its position and speed by radio or mobile data.

Penetration rate p	The share of vehicles that are connected: p = 0.25 means one in four.
Reinforcement learning (RL)	A program that learns by trial and error, getting a reward after each choice.
Max-pressure	A control rule that gives green to the movement with the most waiting traffic and the most room to go to.
SPaT	Signal Phase and Timing: a message telling cars which light is on and when it will change.

The map of today's ideas



1 How to evaluate a new SCATS idea

IN PLAIN WORDS

Testing a new running shoe? Change only the shoe, not the shoe and the track. Race on several days, not one lucky windy day. And time the sprint and the marathon, because a shoe can help one and hurt the other.

- Change one thing through one hook
- Three baselines on identical demand
- Several seeds: mean and spread
- Report delay, stops, bus delay, queue
- Ask what would stop deployment

3 seeds

minimum to see whether a difference is real

Worked example: Is the RL agent worse than SCATS?

$$\text{difference per seed } d_s = \text{delay}_{new} - \text{delay}_{SCATS}$$
$$\text{report mean}(d) \text{ and spread (standard deviation) of } d$$

Mean delay (s/veh) on three unseen seeds. SCATS: 37.91, 35.57, 35.86. RL agent: 39.37, 42.27, 40.27.

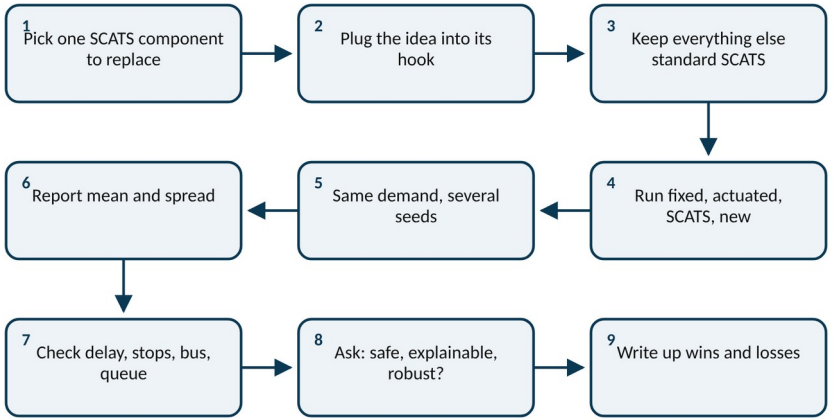
1. Differences: 39.37 − 37.91 = **1.46**, 42.27 − 35.57 = **6.70**, 40.27 − 35.86 = **4.41**.

2. Mean difference: (1.46 + 6.70 + 4.41) ÷ 3 = 12.57 ÷ 3 = **4.19 s**.

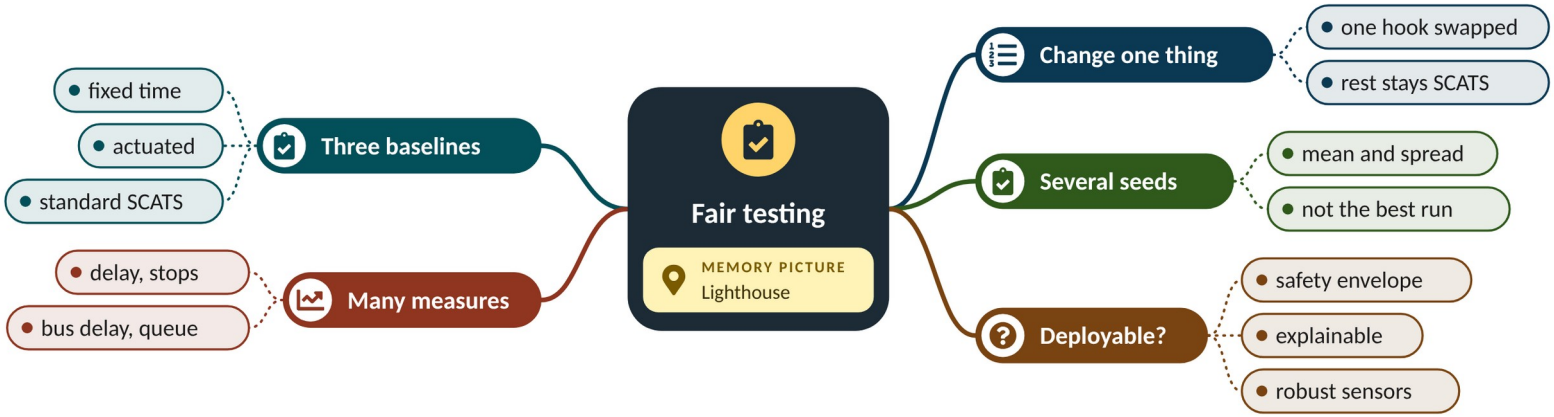
3. Spread (sample standard deviation): about **2.63 s**.

4. All three differences are positive, and the mean is larger than the spread.

The agent is worse by about 4.2 s per vehicle on every seed. One seed alone could have shown anything from 1.5 to 6.7 s.



Remember it: Fair testing



PICTURE THIS · STOP 1: LIGHTHOUSE

At the lighthouse the keeper races four identical toy ships in a bathtub. She changes only one sail on the new ship, then throws three dice to make three different storms, and writes every result on the wall, even the races the new ship lost.

TRAP

Do not tune your new method on the same seeds you use to test it. The RL agent here was trained on other seeds and tested on unseen seeds 7, 8 and 9. Testing on training seeds flatters the method.

2 Degree of saturation from connected vehicles

IN PLAIN WORDS

It is like guessing how many people came to a party from the guests who posted photos. If one guest in four posts, you multiply by four. With only a few posters, one extra photo changes your guess a lot.

- Count connected cars, scale up by 1/p
- Convert to green needed, divide by green
- Noise shrinks like one over root p
- Error 0.65 at 5%, 0.08 at 100%
- Closed loop: 8-21% more delay

0.65

mean DS error with 5% connected vehicles

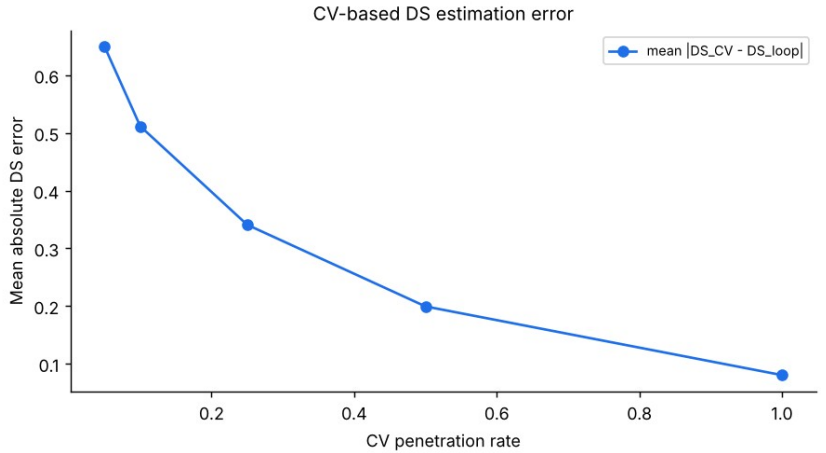
Worked example: One green at 25% penetration

$$\hat{n} = n_{cv} \div p \qquad \hat{g} = \hat{n} \times 3600 \div MF \qquad \hat{DS} = \hat{g} \div g$$
$$\text{error} \propto 1 \div \sqrt{p}$$

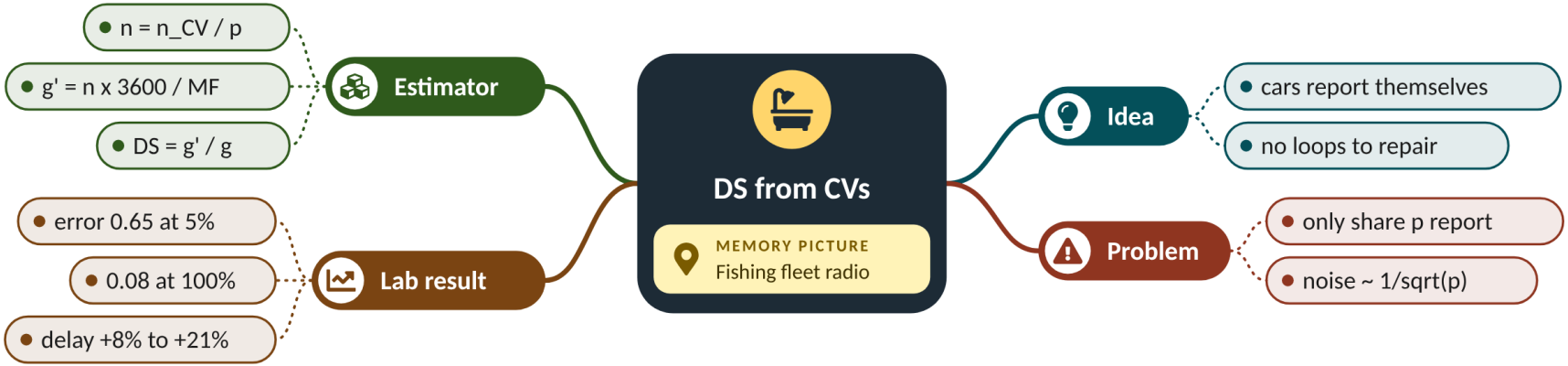
Green g = 30 s. MF = 1,800 veh/h, so a saturated stream needs 3600 ÷ 1800 = 2 s per vehicle. In truth 14 vehicles crossed. 4 of them were connected. p = 0.25.

1. Scale up: $\hat{n} = 4 \div 0.25 =$ **16 vehicles** (truth: 14).
2. Green needed: $\hat{g} = 16 \times 2 =$ **32 s**. Estimated DS = 32 ÷ 30 = **1.07**.
3. If counted perfectly: 14 × 2 = 28 s, DS = 28 ÷ 30 = **0.93**.
4. Spread of the CV count: $\sqrt{(14 \times 0.25 \times 0.75)} =$ **1.62** cars, so $\hat{n}$ wobbles by 1.62 ÷ 0.25 = 6.5 cars, and DS by 6.5 × 2 ÷ 30 = **0.43**.
5. Chance that none of the 14 is connected: $0.75^{14} =$ **0.018**, about 1 green in 56.

One lucky extra CV moved DS from 0.93 to 1.07. Typical wobble is ±0.43, which is why SCATS on 25% CV data cannot match loops green by green.



Remember it: DS from CVs



PICTURE THIS · STOP 2: FISHING FLEET RADIO

On the quay, the harbourmaster counts returning fishing boats by radio. Only one boat in four has a radio, so every 'beep' he hears he multiplies by four on a huge chalkboard. A seagull steps on the radio, one extra beep sounds, and his total leaps by four boats.

TRAP

Do not expect delay to improve smoothly as penetration rises. In closed loop, delay was +11% at 5%, +8% at 25% and +21% at 50%. SCATS's plan voting absorbs some noise, and noisy overestimates lengthen greens, which can help buses and stops by chance.

3 Learning split selection inside the SCATS envelope

IN PLAIN WORDS

Think of a new waiter allowed to recommend only dishes from the chef's menu. He learns from tips which dish suits which table. He cannot invent a dangerous dish, but early on he still recommends worse than the experienced head waiter.

- Agent picks only approved split plans
- Safety limits stay with SCATS
- Tabular Q-learning, reward = fewer halted cars
- Lost by about 4 s per vehicle
- Next: richer state, longer training

+4.2 s

RL delay penalty against SCATS, 3 seeds

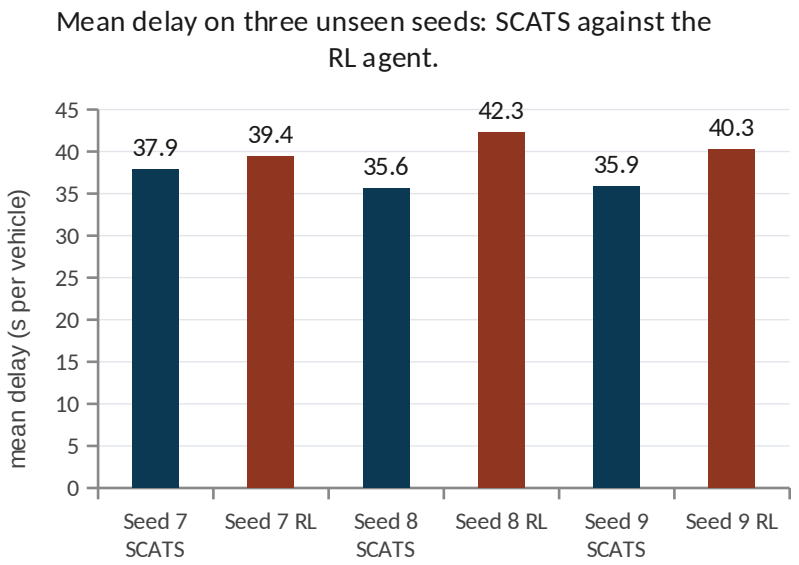
Worked example: One Q-learning update

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[\, r + \gamma \cdot \max Q(s',\cdot) - Q(s,a)\,]$$
$$\text{target} = r + \gamma \cdot \max Q(s',\cdot)$$

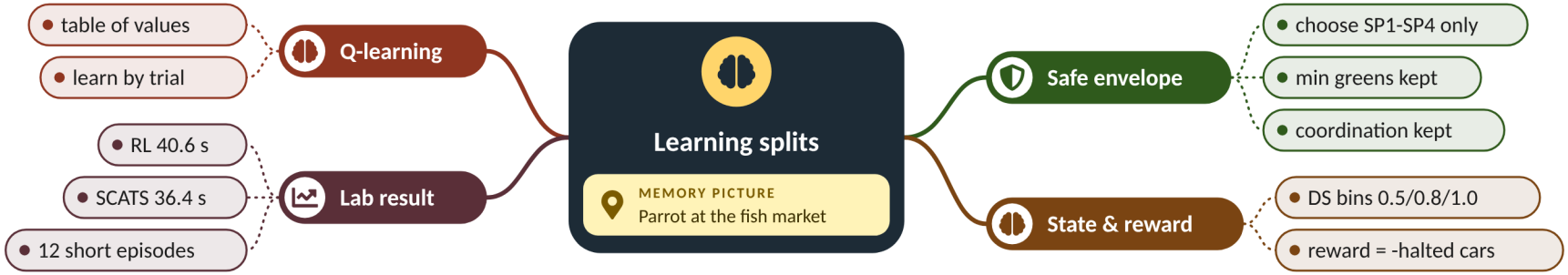
Current Q(s, SP2) = −5. The agent picks SP2 and sees reward r = −4 (four halted cars). The best value in the next state is −3. Learning rate α = 0.1, discount γ = 0.9.

1. Target: −4 + 0.9 × (−3) = −4 − 2.7 = **−6.7**.
2. Surprise (error): −6.7 − (−5) = **−1.7**.
3. Step: 0.1 × (−1.7) = **−0.17**.
4. New Q(s, SP2) = −5 − 0.17 = **−5.17**.

SP2 now looks slightly worse in this state, so the agent will try it a little less. Thousands of such small updates are needed; 12 short episodes give only a few hundred per state.



Remember it: Learning splits



PICTURE THIS · STOP 3: PARROT AT THE FISH MARKET

In the fish market a parrot sits in a cage holding exactly four menu cards. It may only shout a dish from those cards. When the queue at the counter shrinks it gets a sardine. It still shouts the wrong card so often that its queue is four people longer than the old fishmonger's.

TRAP

A losing result is not a useless result. The agent lost because its state is coarse, its reward is noisy, it trained for only 12 episodes, and SCATS's DS rule is a strong, well-tuned baseline. Each reason is a concrete next experiment.

4 SCATS and max-pressure

IN PLAIN WORDS

Think of doors between crowded rooms. Open the door where many people wait on your side and few on the other side. Opening a door into an already packed room just moves the crush.

- Pressure: waiting queue minus downstream load
- Pure max-pressure breaks green waves
- Hybrid keeps cycle and offsets, blends greens
- Corridor: SCATS 36.9, hybrid 40.2 s
- Next: test on a congested grid

57.4 s

bus delay under pure max-pressure (SCATS 29.7 s)

Worked example: Blending 70 s of green between two phases

$$\text{pressure}_p = \Sigma\,(\,\text{queue}_{\text{upstream}} - \tfrac{1}{2}\cdot\text{load}_{\text{downstream}}\,)$$

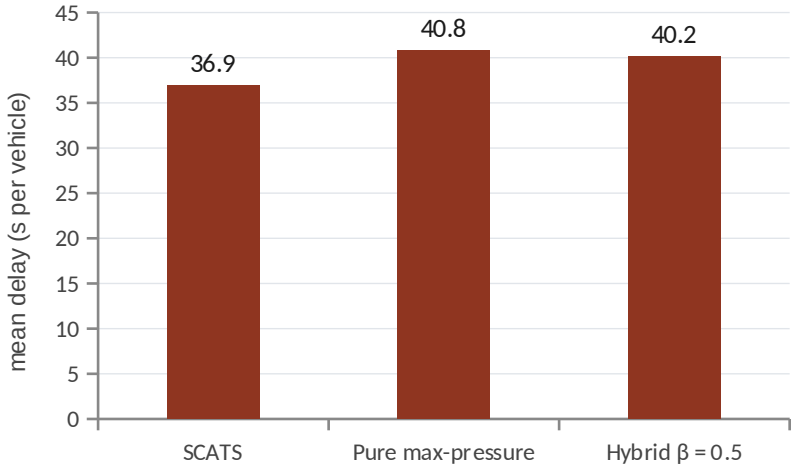
$$g_p = (1-\beta)\cdot g_{\text{SCATS},p} + \beta\cdot\text{avail}\cdot\text{pressure}_p \div \Sigma\,\text{pressure}$$

Phase A: 12 cars queued, 16 on its exit road. Phase B: 8 queued, 2 on its exit. 70 s of green to share. SCATS's own split gives A 40 s, B 30 s. β = 0.5.

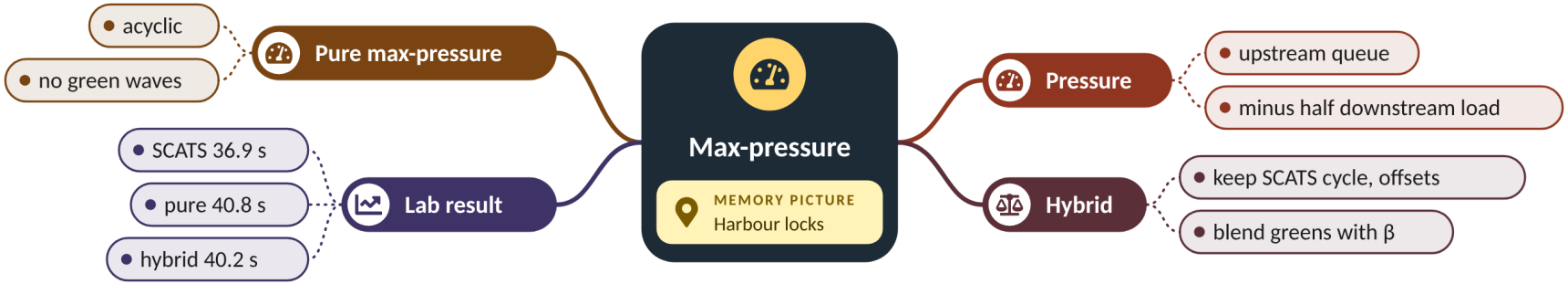
1. Pressure A = 12 − ½ × 16 = **4**. Pressure B = 8 − ½ × 2 = **7**. Total 11.
2. Pressure greens: A = 70 × 4 ÷ 11 = **25.5 s**; B = 70 × 7 ÷ 11 = **44.5 s**.
3. Blend A: 0.5 × 40 + 0.5 × 25.5 = **32.7 s**.
4. Blend B: 0.5 × 30 + 0.5 × 44.5 = **37.3 s**. Check: 32.7 + 37.3 = 70 s.

Pressure pulls about 7 s away from A, whose exit is filling up. On a congested grid that prevents spill-back; on an uncongested corridor it just disturbs a good SCATS split.

Mean delay on the corridor: SCATS, pure max-pressure and the hybrid.



Remember it: Max-pressure



PICTURE THIS · STOP 4: HARBOUR LOCKS

At the harbour locks two gates hold back boats. A pressure gauge on each gate shows boats waiting minus half the boats crammed in the basin beyond. The lock keeper turns a dial marked β halfway, mixing his old timetable with the gauges. Meanwhile the water bus, ignored, waits so long it sinks to its chimney.

TRAP

Max-pressure being 'throughput-optimal' does not mean 'lowest delay everywhere'. The proof assumes idealised queues and no platoons. Test it where it should shine: a grid with spill-back and turning conflicts.

5 Resilience to detector faults

IN PLAIN WORDS

Imagine a smoke alarm stuck silent and another that never stops beeping. The silent one says all is calm; the beeping one says the house is always on fire. A sensible person notices 'this alarm has beeped for two days' and ignores it until it is repaired.

- Stuck-off: DS near zero, early gap-out
- Stuck-on: DS near one, runs to max
- 30% faulty loops: delay up 60%
- Simple alarms recover about 83%
- Exclude or impute: both work here

+60%

delay increase with 30% of loops faulty and no alarms

Worked example: What two broken loops report on a 30 s green

$$DS = (g - (T - n \cdot h)) \div g$$
$$\text{recovered} = (\text{raw} - \text{fixed}) \div (\text{raw} - \text{healthy})$$

Green g = 30 s, h = 1.6 s. Healthy loop 1: 6 vehicles, unoccupied for 15.6 s. Phase DS = highest DS among its loops.

1. Healthy loop: g' = 30 − (15.6 − 6 × 1.6) = 30 − 6.0 = 24 s; DS = 24 ÷ 30 = **0.80**.

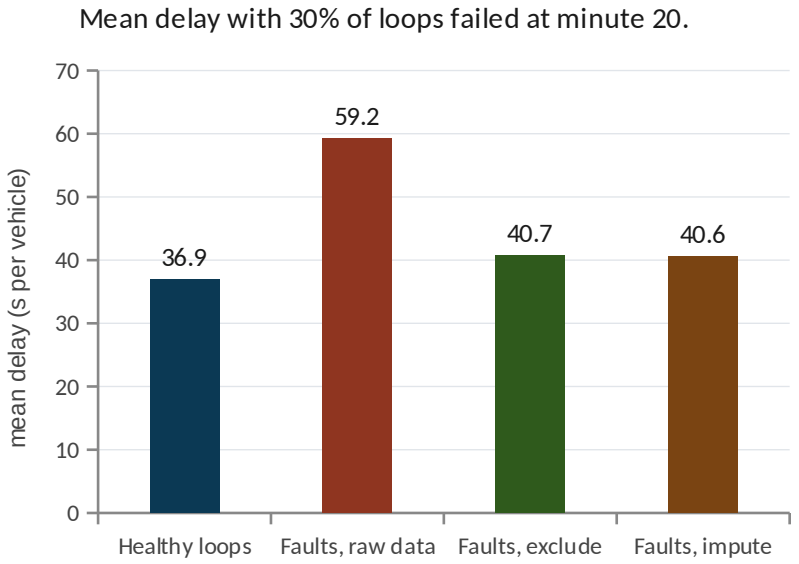
2. Stuck-off loop: n = 0, T = 30, so DS = (30 − 30) ÷ 30 = **0**. The highest is still 0.80, but if every loop of a phase is stuck off, DS = 0.

3. Stuck-on loop: T = 0, so DS = (30 − 0) ÷ 30 = **1.00**. The phase reads 1.00, wrongly saturated.

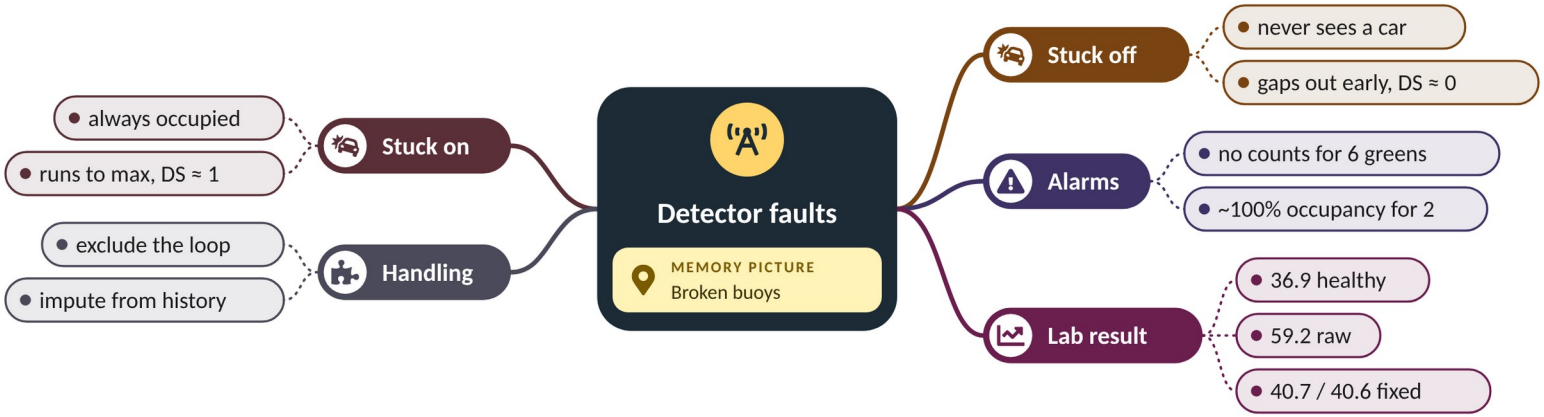
4. Exclude the alarmed stuck-on loop: phase DS = **0.80** again.

5. Lab recovery: (59.17 − 40.70) ÷ (59.17 − 36.94) = 18.47 ÷ 22.23 = **0.83**.

One stuck-on loop is enough to make a lightly used phase look full. Excluding it restores the right DS, and in the lab such handling recovered 83% of the lost performance.



Remember it: Detector faults



PICTURE THIS · STOP 5: BROKEN BUOYS

In the bay floats a row of bell buoys. One bell is silent, wrapped in seaweed. Another rings non-stop, driving the gulls mad. A diver swims up and ties a red alarm flag to each, and the ringing one is muted. A delay balloon tied to the pier swells from 37 to 59, then shrinks back to 41.

TRAP

Do not assume a faulty loop always shows up as 'zero'. A stuck-on loop is worse for the strategic layer: it looks like heavy traffic, so it can push the whole subsystem's cycle length up.

6 Predicting signal timing for connected vehicles

IN PLAIN WORDS

It is like a bus app predicting arrival. A timetable is easy to read but the driver may hurry or wait. The best guess is often the driver's own plan for today, not what the bus did yesterday.

- SPaT tells cars when lights change
- Adaptive control re-plans, so it's uncertain
- Planned green best: 2.63 s error
- History alone did worse
- Next: live updates and confidence bands

2.63 s

mean error of the planned-green predictor

Worked example: Two predictors on four greens

$$\text{MAE} = (|\text{actual}_1 - \text{predicted}_1| + \ldots + |\text{actual}_n - \text{predicted}_n|) \div n$$

Actual greens: 31, 26, 34, 29 s. Planned: 32, 30, 33, 30 s. Persistence (last cycle's green): 28, 31, 26, 34 s.

1. Planned errors: |31 − 32| = 1, |26 − 30| = 4, |34 − 33| = 1, |29 − 30| = 1.

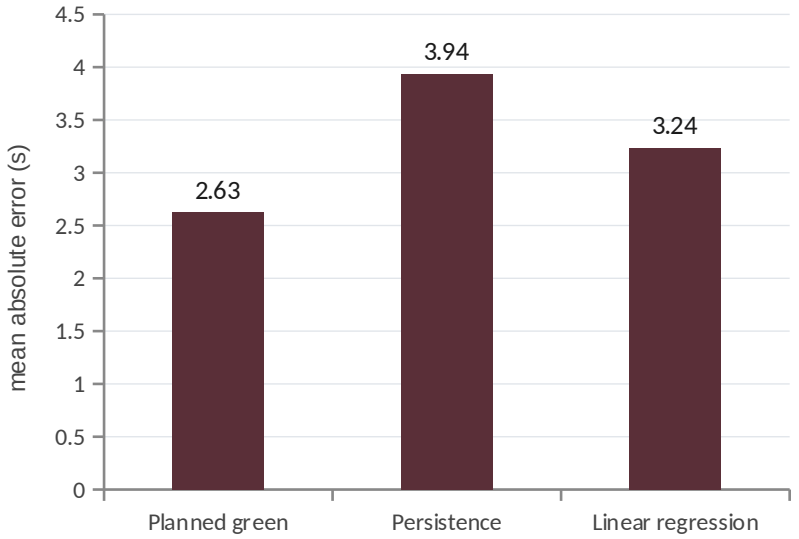
2. Planned MAE = (1 + 4 + 1 + 1) ÷ 4 = 7 ÷ 4 = **1.75 s**.

3. Persistence errors: |31 − 28| = 3, |26 − 31| = 5, |34 − 26| = 8, |29 − 34| = 5.

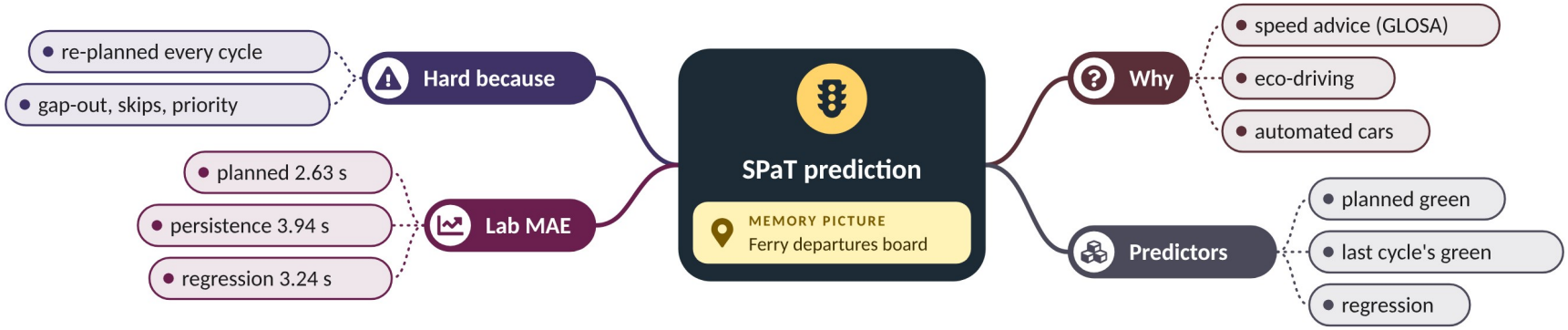
4. Persistence MAE = (3 + 5 + 8 + 5) ÷ 4 = 21 ÷ 4 = **5.25 s**.

The plan wins because it already includes this cycle's decision. Persistence is always one cycle behind. The same ranking appeared in the lab (2.63 s against 3.94 s). Numbers here are illustrative.

Green-duration prediction error on the SUMO corridor.



Remember it: SPaT prediction



PICTURE THIS · STOP 6: FERRY DEPARTURES BOARD

At the ferry terminal a departures board shows the next sailing with a small error, about two and a half minutes. Beside it an old sailor shouts yesterday's times and is off by four. A third man with a calculator lands in between. A ferry suddenly leaves early because nobody is boarding.

TRAP

A low average error can hide large misses. A green that gaps out 10 s early can catch a car that trusted the prediction. Publish a confidence band, not just one number, and update it every second.

Five research directions at a glance

Direction	SCATS part replaced	Lab result	Beat SCATS?
CV-based DS	ds_source	delay +8% to +21%	no
RL split selection	split_selector	40.6 vs 36.4 s	no
Max-pressure hybrid	green_allocator	40.2 vs 36.9 s	no (corridor)
Fault alarms	sensor_filter / fault_handling	59.2 → 40.7 s	recovers 83% of fault loss
SPaT prediction	none (prediction only)	planned MAE 2.63 s	best of three predictors

Things beginners mix up

This…	…is not this	The fact that tells them apart
Throughput-optimal	**Lowest delay**	Max-pressure's proof is about carrying the most traffic under ideal queues, not delay with platoons.
Stuck-off loop	**Stuck-on loop**	Stuck-off reads DS 0 and gaps out early; stuck-on reads DS 1 and runs to maximum.
Shadow mode	**Closed loop**	Shadow mode only compares estimates on the same traffic; closed loop lets the estimate drive the lights.
Penetration p	**Accuracy**	Error falls like $1/\sqrt{p}$: four times the penetration halves the error.
Training seeds	**Test seeds**	Evaluate on seeds the method never trained on (here 7, 8, 9).

Check yourself (1 of 2)

p = 0.2, green 30 s, MF 1,800 veh/h, 3 CVs crossed. Estimated DS?

A. 0.20

B. 0.50

C. 1.00

D. 1.50

Phase A: queue 9, downstream 10. Phase B: queue 5, downstream 0. Which has the higher pressure?

A. A (4)

B. B (5)

C. Equal

D. Cannot tell

Answers are in the speaker notes and in the study guide's answer key.

Check yourself (2 of 2)

Healthy 36.9 s, raw faults 59.2 s, with alarms 40.7 s. What share of the fault damage was recovered?

A. about 40%

B. about 60%

C. about 83%

D. 100%

A GLOSA app uses the planned green (MAE 2.63 s). What is the main remaining risk?

A. Average error too large

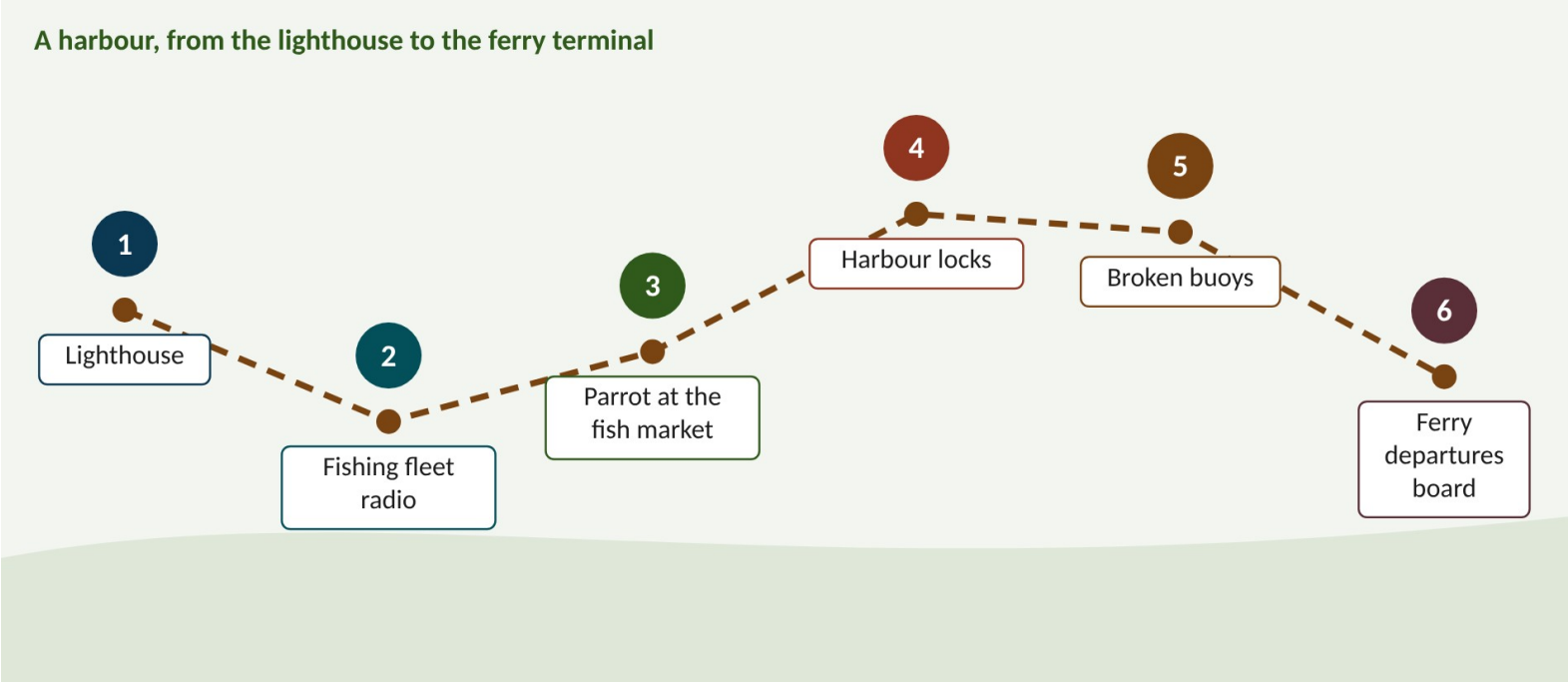
B. Rare large misses from gap-out or priority calls

C. Cars cannot receive SPaT

D. Planned greens are random

Answers are in the speaker notes and in the study guide's answer key.

Walk the memory route



1 Lighthouse = Fair testing · 2 Fishing fleet radio = DS from CVs · 3 Parrot at the fish market = Learning splits · 4 Harbour locks = Max-pressure · 5 Broken buoys = Detector faults · 6 Ferry departures board = SPaT prediction

Numbers to remember

- RL vs SCATS on 3 seeds: 40.6 vs 36.4 s (+4.2 s)
- CV DS error: 0.65 (5%), 0.34 (25%), 0.08 (100%)
- CV closed loop: delay +8% to +21%
- Corridor: SCATS 36.9, pure MP 40.8, hybrid 40.2 s
- Bus delay: SCATS 29.7, pure MP 57.4 s
- Faults: 36.9 → 59.2 → 40.7 / 40.6 s
- SPaT MAE: planned 2.63, regression 3.24, persistence 3.94 s

Keep going

Study guide: SCATS_Mastery_Study_Guide.pdf · Interactive labs and SUMO code: smrahmanbd.github.io/scats-learning-lab